Mumbai University Bachelor's of Computer Engineering(Syllabus)

Semester I

| Course code | Course name | Module Name |
|---|---|---|
| BSC101 | AppliedMathematics-I | 1. Complex Numbers<br>2. Hyperbolic Functions & Logarithms of Complex Numbers<br>3. Partial Differentiation<br>4. Applications of Partial Differentiation and Successive Differentiation.<br>5. Matrices<br>6. Numerical Solutions of Transcendental Equations and System of Linear Equations and Expansion of Function. |
| BSC102 | Applied Physics | 1. Lasers<br>2.Fibre Optics<br>3. Interfenence In Thin Films<br>4. Electrodynamics<br>5.Quantum Physics<br>6. Basics Of Semiconductor Physics |
| BSC103 | Applied Chemistry | 1. Fuels and Combustion<br>2.Corrosion<br>3.Alloys<br>4. Introduction to Polymers<br>5. Introduction to Composites<br>6. Green Chemistry for sustainable development |
| ESC101 | Engineering Mechanics | 1. System of Forces<br>2.Centroid<br>3. Equilibrium of Force system and Friction<br>4. Kinematics of particle and rigid bodies<br>5.Kinetics of Particle<br>6. Introduction to Robot Kinematics |
| ESC102 | BasicElectrical&Electronics Engineering | 1. DC Circuits<br>2. AC Circuits<br>3. Single Phase Transformer<br>4. Electrical Machines<br>5. Special Purpose Diodes<br>6. Introduction to Transistors |
| AEC101 | Professional and Communication Ethics | 1. Fundamentals of Communication<br>2. Developing Basic Listening Skills<br>3. Developing Basic Speaking Skills<br>4. Developing Basic Reading Skills<br>5. Developing Basic Writing Skills |

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| | | 6. Ethical and Managerial Skills for Engineers |
| VSEC101 | Engineering Workshop-I | 1. Fitting(Compulsory)<br>2. HardwareandNetworking<br>3. Welding<br>4. Plumbing<br>5. MachineShop |
| VSEC102 | C Programming | 1. Fundamentals of C Programming<br>2.Control Structures<br>3. Functions and Parameter<br>4. Arrays , String Structure<br>5.Pointers<br>6.Files |
| CC101 | Inductioncum Universal Human Values | 1. Introduction - Need, Basic Guidelines, Content and Process for Value Education<br>2. Understanding Harmony in the Human Being<br>3. Understanding Harmony in the Family<br>4. Understanding Harmony in the Society<br>5. Understanding Harmony in the Nature and Existence - Whole existence as Coexistence<br>6. Implications of the Holistic Understanding of Harmony on Professional Ethics |

Semester II (syllabus)

| Course Code | Course Name | Module Name |
|---|---|---|
| BSC201 | Applied Mathematics– II | 1. Differential Equations of First Order and First Degree<br>2. Linear Differential Equations With Constant Coefficients of Higher Order<br>3. Beta and Gamma Function, Differentiation under Integral sign<br>4. Multiple Integration- I Pre-requisite: Tracing of curves<br>5. Multiple Integration- II<br>6. Numerical solution of ordinary differential equations of first order and first degree, and , Numerical Integration |
| BSC202X | Elective Physics | |
| BSC2021 | Physics for Emerging Fields | 1. solar Energy<br>2. OPTICAL Imaging<br>3. Micro Electro- Mechanical Systems<br>4. Fuel Cell |

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| | | 5. Energy harvesting<br>6. Nanocomputing |
| BSC2022 | Semiconductor Physics | 1. Basics of Semiconductors<br>2. Junction diode<br>3. Important Diodes<br>4. Bipolar Junction Transistors<br>5. Field Effect Transistors<br>6. NanoTechnology |
| BSC2023 | Physics of Measurements and Sensors | 1. Introduction<br>2. Measurements by light – Wave Interference<br>3. Transducers<br>4. Solid state sensors<br>5. Temperature and its measurements<br>6. Nanotechnology |
| BSC203X | Elective Chemistry | |
| BSC2031 | Engineering Materials | 1.Alloys<br>2.Ceramics<br>3.Composites<br>4. Plastics and Elastomers<br>5. Advanced Polymers<br>6.Nano Materials |
| BSC2032 | Environmental Chemistry and Non-conventional energy sources | 1. Air Pollution and Atmospheric Chemistry<br>2. Water & Waste water Treatment and Management<br>3. Solid and Hazardous Waste Management<br>4. Introduction to non conventional (Renewable) energy source<br>5. Non conventional Energy sources<br>6. Sustainable Practices |
| BSC2033 | Introduction to Computational Chemistry | 1. Introduction to Computational Chemistry<br>2. Tools of Computational Chemistry<br>3. Basics of Quantum mechanics<br>4. Molecular mechanics<br>5. Molecular Structure and Bonding<br>6. Computational Methods in Quantum Chemistry |
| ESC201 | EngineeringGraphics | 1. Introduction to Engineering Drawing<br>2. Projections of Points, Lines and Planes<br>3. Projections of Solids<br>4. SectionsofSoli ds and Development of Surfaces<br>5. Orthographic Projections<br>6.Isometric Views |
| PCC201X | Program Core Course (Data Structure ) | 1. Introduction<br>2.Stacks |

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| | | 3.Queue<br>4.Linked List<br>5.Tree<br>6. Applications of Data Structures |
| CC201 | Social Science & Community Services | (Group Activities) |
| IKS201 | IndianknowledgeSystem | 1. Introduction to the Indian Knowledge System (I.K.S.)<br>2. Development of Scientific Thoughts in Ancient India<br>3. Development of Arts & Culture in India<br>4. Good Governance in Ancient India<br>5. Contribution of Indian Scientist & Nobel Laureates<br>6. Sustainable Practices in |
| VSEC201 | Engineering Workshop-II | 1. Carpentry(Compulsory)<br>2. Basic Electrical work shop:(Compulsory)<br>3. Masonry<br>4. Sheet metal working and Brazing<br>5. Forging (Smithy) |
| VSEC202 | Python Programming | 1. Introduction to Python<br>2. Control Flow and Functions<br>3. File Handling, Packaging, and Debugging<br>4. Object-Oriented Programming (OOP) in Python<br>5. Advanced Python Concepts<br>6. Python Libraries |